

Smart Home Project

Introduction

In this project, you are supposed to design and implement an **energy management algorithm** for a smart household equipped with the following features:

- 20-25 electrical devices,
- A photovoltaic (PV) system,
- A stationary home battery,
- An electric vehicle (EV) with Vehicle-to-Grid (V2G) capability,
- A three-period time-of-use (TOU) electricity tariff.

The aim of the project is to **reduce household electricity costs and peak demand** while respecting **comfort constraints** and handling **uncertainties** in renewable generation and device usage.

This project focuses on a **rule-based decision logic** (e.g., decision trees, priority rules) rather than mathematical optimization. Students should implement simple scheduling rules and explain the reasoning behind them.

Scenario Description

- **Tariff:** Three-time periods (Day, Peak, Night), with different £/kWh rates.
- **Devices:** 20 or 25 household appliances, each with parameters:
 - Nominal power,
 - Duration of operation,
 - Earliest start,
 - Latest end,
 - Flexibility (priority).
- **PV system:** Hourly generation profile based on irradiance.
- **Battery:** Stationary storage with defined capacity, SOC limits, and round-trip efficiency.
- **EV (V2G):**
 - Battery capacity,
 - Max charge/discharge power,
 - Efficiency,

- Start SOC,
 - Required SOC at morning departure,
 - **Constraints:** Comfort constraints (e.g., temperature limits, appliance usage times) must be respected. Some constraints are **hard**(mandatory), while others are **soft** (can be violated with penalties). These will influence scheduling decisions.
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To-Do List

1. Data Preparation

- Define the 20 or 25 devices with their parameters.
- Categorize devices into priority groups (critical, shiftable, optional).
- Prepare typical **winter** and **summer** load profiles.
- Collect meteorological data for PV (and optionally wind).

2. Household Load Profile

- Build the baseline daily load curve.
- Identify peak demand and set a threshold limit.

3. Renewable Generation

- Calculate hourly PV generation.
- Apply priorities:
 1. Local consumption,
 2. EV/battery charging,
 3. Grid export (optional).

4. Stationary Battery

- Define capacity (~50% of peak load).
- Apply charge/discharge power limits.
- SOC constraints: start, min, max.
- Battery operates mainly during **peak hours (17:00–22:00)**.

5. Electric Vehicle (V2G)

- Define EV parameters and availability.
- Model EV as flexible storage.

- Update SOC considering efficiencies.
- Constraint: morning SOC $\geq$ required target.

6. Constraints

Define the constraints as hard and soft constraints.

- **Hard Constraints (Mandatory)**

Hard constraints must **always be satisfied**. If they are violated, the algorithm is considered incorrect, or a significant grade penalty is applied. They represent physical limits, safety requirements, or essential user needs.

Examples:

- EV state of charge (SOC) $\geq$ 70% by 07:00 (departure time).
- EV SOC must never fall below 20%.
- Oven must operate strictly between 18:00–20:00.
- Washing machine must complete its cycle before 22:00.
- Indoor temperature must never exceed 25 °C (summer) or drop below 20 °C (winter).

- **Soft Constraints (Desirable but Flexible)**

Soft constraints are **comfort-related or user-preference-based**. They can be violated, **if necessary**, but each violation should incur a penalty or be justified in the report. They allow students to balance cost, peak demand, and comfort trade-offs.

Examples:

- Indoor temperature comfort band: ideally 21–24 °C. Slight deviations (e.g., 24.5 °C) are allowed but penalized.
- Noise restrictions: avoid running noisy appliances (e.g., vacuum, washing machine) after 22:00. If done, note the comfort impact.
- EV charging time: ideally completed overnight, but minor delays are acceptable if peak demand is reduced.
- Dishwasher: ideally finishes by 22:00, but 23:00 is acceptable if justified.

Please follow these guidelines while working on the project:

- Explicitly list all hard and soft constraints used in your algorithm.
- Ensure hard constraints are always enforced in your decision tree logic.

- If a soft constraint is violated, explain why and quantify the impact.
- Discuss trade-offs: e.g., “Reducing peak demand by 20% required a temporary temperature rise to 24.5 °C.”

7. Load Shifting Algorithm

- If total demand exceeds the defined threshold:
 - Postpone or reschedule non-priority devices within their allowed time windows.
- Use EV and battery discharging to reduce peaks if they are available and within SOC limits.
- Adjust AC setpoint within the comfort band to reduce consumption during peak hours.
- Prioritize the use of PV generation for local consumption and charging EV/battery.
- Implement simple decision rules rather than mathematical optimization:
 - Example: “If load > threshold → defer shiftable devices → adjust AC setpoint → discharge EV → discharge battery.”
- Test and compare system behavior under different scenarios:
 - PV generation variability,
 - Device usage uncertainty,
 - EV arrival/departure time deviations.

8. Outputs

- Graphs:
 - Daily load curve (before vs. after shifting),
 - PV generation,
 - Battery SOC,
 - EV SOC,
 - Grid import/export.
- Calculations:
 - Daily electricity cost,
 - Cost comparison: with vs. without shifting,
 - Separate analysis for winter and summer.
- Comforts:

- Number of comfort violations (soft constraint breaches)
- Temperature deviation hours (how many hours outside 21–24 °C)
- Appliance deadline delays

9. Reporting

- Explain algorithm and scheduling approach.
- Present input parameters (tariff, devices, PV, EV, battery).
- Show results for different uncertainty scenarios.
- Discuss trade-offs: cost, comfort, flexibility, peak reduction.

Deliverables

1. **Code** (Python, MATLAB, or equivalent): clean, commented, reproducible.
2. **Input files**: parameters (CSV/JSON).
3. **Outputs**:
 - Plots of load curves, SOC, grid power, etc.
 - Cost and performance metrics.
4. **Report (max 5 pages)**:
 - Method description, results, comparisons, discussion.
5. **Short oral defense** (optional): explain your choices and answer questions.

Evaluation Criteria

- **Technical correctness (50%)**: SOC updates, constraints satisfied, load shifting logic.
- **Comfort compliance (10%)**:
 - Hard constraints respected (✓/X).
 - Soft constraints handled reasonably (penalty-based or explained trade-off).
- **Analysis & reporting (35%)**: clarity of graphs, comparisons, scenario discussion.
- **Code quality (5%)**: readability, reproducibility.

Additional Notes on Comfort Constraints

Air Conditioner (AC) Setpoint vs. Electrical Demand

- Assume each **+1 °C increase in cooling setpoint reduces average AC electrical power by 5–10%**.
- It must be **chosen one value** (5%, 7%, or 10%) and justify the choice.
- Select **a single value** (5%, 7%, or 10%) for the reduction and provide a justification for your choice.
- Conduct **a sensitivity analysis** using at least two different values for the reduction percentage.

Example for a 3.0 kW AC:

- +1 °C at 5% → 2.85 kW
- +1 °C at 7% → 2.79 kW
- +1 °C at 10% → 2.70 kW

Implementation formula:

Comfort band rule: indoor temperature must remain within **21–24 °C**.

Electric Vehicle (EV) Charging & Discharging

- **Capacity:** 30–50 kWh
- **Max charge power ($P_{\text{chg_max}}$):** 3.5–7.0 kW.
- **Max discharge power ($P_{\text{dischg_max}}$):** $\leq P_{\text{chg_max}}$ (e.g., 3.5–5.0 kW).
- **Efficiency:** $\eta_{\text{chg}} = \eta_{\text{dischg}} = 0.9$.
- **SOC constraints:**
 - $\text{min_SOC} = 20\%$ (do not discharge below this),
 - required morning SOC = 70% (by 07:00),
 - reserve margin = 10% (for unexpected trips).

Rule-based scheduling:

1. **Discharge** if:
 - $\text{load} > \text{threshold}$,
 - EV connected,
 - $\text{SOC} > \text{min_SOC} + \text{margin}$.
2. **Charge** if:
 - PV surplus exists, or

- Tariff period = Night, and SOC < required.
3. **Do not discharge** if:
- SOC < min_SOC, or
 - Next departure would risk SOC < required.
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Student Tasks

- Define device priority rules and explain them in the report.
- Select **AC savings percentage (5–10%)** and run at least two sensitivity cases.
- Define **SOC thresholds** for EV and battery.
- Implement the decision-tree rules.
- Present **before/after load curves**, SOC plots, and comfort analysis.